Additional file 2. Cohort audit

This file supports the main-text section *Cohort assembly and audit* and Table 1 of the main text. Public disease series are frequently supersets, re-analyses or re-runs of one another. Counting such a series as an independent cohort inflates every cross-cohort agreement statistic, because the agreement is partly between a patient and themselves.

Procedure

For every candidate series we read the sample-level metadata from the GEO series matrix, extracted the subject identifier where the submitters recorded one, and compared identifier sets across series. Where identifiers were absent we compared sample titles, platform, and the exact per-sample value vectors. A series was excluded when its subjects were already present in a series we retained.

Excluded series (3)

Accession	Superseded by	Evidence
GSE30122	GSE30528, GSE30529	duplicates GSE30528 + GSE30529 and re-runs 25 of the same controls
GSE47183	GSE104948	DN subject IDs identical to GSE104948
GSE99340	GSE104948, GSE104954	ERCB superset of GSE104948 / GSE104954; ~51 samples are cell lines

Residual overlap that could not be removed

64 subjects appear in more than one retained series. These are not duplicate cohorts but the same ERCB patients profiled in two compartments, which is what makes the paired glomerular and tubulointerstitial analyses possible. The benchmark treats the two compartments as separate datasets, so the leave-one-dataset-out folds are not fully independent for those subjects; this is stated in the limitations of the main text.

Series A	Series B	Shared subjects	Subjects in A	Subjects in B
GSE104948	GSE104954	55	196	195
GSE30528	GSE30529	9	22	22

Full audit table

Accession	Omics	Samples loaded	Status	Note
GSE30122	transcriptomics	69	excluded	duplicates GSE30528 + GSE30529 and re-runs 25 of the same controls
GSE47183	transcriptomics	122	excluded	DN subject IDs identical to GSE104948

Accession	Omics	Samples loaded	Status	Note
GSE99340	transcriptomics	413	excluded	ERCB superset of GSE104948/GSE104954; ~51 samples are cell lines
GSE1009	transcriptomics	6	retained	
GSE104948	transcriptomics	196	retained	
GSE104954	transcriptomics	195	retained	
GSE111154	transcriptomics	8	retained	
GSE131882	transcriptomics		retained	
GSE142025	transcriptomics	36	retained	
GSE142153	transcriptomics	40	retained	
GSE175759	transcriptomics		retained	
GSE20602	transcriptomics	18	retained	
GSE30528	transcriptomics	22	retained	
GSE30529	transcriptomics	22	retained	
GSE96804	transcriptomics	61	retained	
ST000691	metabolomics		retained	downloaded, not yet analysed
ST002145	metabolomics		retained	downloaded, not yet analysed
ST004442	metabolomics		retained	downloaded, not yet analysed
ST004483	metabolomics		retained	downloaded, not yet analysed

Samples loaded is the number of samples parsed into the project database. It is zero for assets that were downloaded and inspected but not parsed into the sample table, such as the single-cell objects and most metabolomics studies, which are analysed from their own files.